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ARCHIVE HISTORIQUE - T1'-T4 argument invalidated

Editorial notice - 20 October 2026

The historical T1'-T4 chain does not prove the Riemann hypothesis. The public erratum of 13 October 2026 identifies errors in T2 (weighted Gram-matrix bound) and T3 (wave-packet normalization and capture argument).

Earlier titles, abstracts and claims of complete or journal-ready proofs are historical claims withdrawn as proof arguments. This withdrawal does not assert that every mathematical object cited is false.

This editorial page has been added to the historical PDF. All original pages follow with their content unchanged. This is neither a recompilation nor a new mathematical result.

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Public T1'-T4 erratum - 13 October 2026

A Modular Analytic Framework for the Riemann Hypothesis (T1′–T4)

Proof-only, journal-ready, arXiv-compatible

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Abstract

Proof framework (T1′–T4). This arXiv v1 tarball includes full proofs for T2, T3, and the appendix.

1 T2 — Weighted frame bound (full)

1.1 Weighted frame bound (T2): full proof

1.1.1 Statement

Let $\phi \in \mathcal{PW}$ be even, $\operatorname{supp}\hat{\phi} \subset [-A, A]$, $\hat{\phi}(0) = 1$.
Let $\Xi_A = \{\xi_{p,k} = k \log p \le A\}$ and weights $w_{p,k} = \frac{\log p}{p^{k/2}}$.
Define $G^{(w)}_{\alpha\beta} = \sqrt{w_\alpha}\, \hat{\phi}(\xi_\alpha - \xi_\beta)\, \sqrt{w_\beta}$.

Theorem 1.1. *There exist $A_0 \ge 1$ and $\Theta_w(A) \in (0,1)$ for $A \ge A_0$ such that $\lambda_{\min}(G^{(w)}) \ge 1 - \Theta_w(A) =: C^{(w)}_{\mathrm{fr}}(A) > 0$.*

1.1.2 Schur/Gershgorin

Let $R = G^{(w)} - I$. Schur's test gives

$$\|R\|_{2\to 2} \le \Big(\sup_\alpha \sum_\beta |R_{\alpha\beta}|\Big)^{1/2} \Big(\sup_\beta \sum_\alpha |R_{\alpha\beta}|\Big)^{1/2}.$$

Since R is symmetric,

$$\|R\|_{2\to 2} \le \Theta_w(A) := \sup_\alpha \sum_{\beta \ne \alpha} \sqrt{w_\alpha w_\beta}\, |\hat{\phi}(\xi_\alpha - \xi_\beta)|.$$

Hence $\lambda_{\min}(G^{(w)}) \ge 1 - \Theta_w(A)$.

1.1.3 Locality and summability

Fix $\alpha = (p,k)$. Band-limitation implies $\hat{\phi}(\xi_\alpha - \xi_\beta) = 0$ unless $|k \log p - \ell \log q| \le A$. Split $\sum_{\beta \ne \alpha} = \sum_{q=p} + \sum_{q \ne p}$.

Same prime $q = p$. Then $|\ell - k| \le \lfloor A/\log p \rfloor$ and $\sqrt{w_{p^k} w_{p^\ell}} = (\log p)/p^{(k+\ell)/4} \le (\log p)/p^{k/2} \cdot p^{-(\ell-k)/4}$, whence

$$\sum_{\substack{\ell \ne k \\ |k-\ell| \le A/\log p}} \sqrt{w_{p^k} w_{p^\ell}}\, |\hat{\phi}(\xi_{p,k} - \xi_{p,\ell})| \;\le\; \frac{C_1}{p^{k/2 - 1/4}}.$$

Different primes $q \neq p$ (refined). Locality forces $q^\ell \in [p^k e^{-A}, p^k e^A]$, so $q \leq p^k e^A$ and the number of admissible primes is $\ll p^k e^A / \log(p^k e^A)$ (PNT). For each q, admissible ℓ lie in an interval of length $\ll A/\log q$. Using $\sqrt{w_{p^k} w_{q^\ell}} \leq \sqrt{\log p \log q}\, p^{-k/4} q^{-\ell/4}$ and $\sum_{q \leq X} \frac{\sqrt{\log q}}{q^{1/4}} = O(X^{3/4})$,

$$\sum_{q \neq p} \sum_{\ell \text{ admissible}} \sqrt{w_{p^k} w_{q^\ell}}\, |\hat{\phi}(\xi_{p,k} - \xi_{q,\ell})| \;\leq\; \frac{C_2 A}{p^{k/4}} (p^k e^A)^{3/8} \;=\; C_3\, A\, p^{k/8}\, e^{3A/8}.$$

Weighted by $w_\alpha = (\log p)/p^{k/2}$ yields $\ll A e^{3A/8}\, p^{-3k/8}$, geometrically small in k.

Conclusion. Thus

$$\sum_{\beta \neq \alpha} \sqrt{w_\alpha w_\beta}\, |\hat{\phi}(\xi_\alpha - \xi_\beta)| \;\leq\; \frac{C_1}{p^{k/2-1/4}} + \frac{C_3 A e^{3A/8}}{p^{3k/8}},$$

so $\Theta_w(A) < 1$ for $A \geq A_0$ and $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A) > 0$.

2 T3 — Wave–packet localization (full)

2.1 Wave–packet localization (T3): full proof

Assume Theorem 1.1 and the explicit formula (Appendix). Let $\gamma_0 \in (0, A)$ and suppose a nontrivial zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 \neq \frac{1}{2}$.

2.1.1 Wave–packet construction

Pick $b \in C_c^\infty([-1,1])$, even, $\int b = 1$. For $T \geq 1$ set

$$\hat{\phi}_T(\gamma) = \tfrac{1}{2}\Big(b(\sqrt{T}(\gamma - \gamma_0)) + b(\sqrt{T}(\gamma + \gamma_0))\Big).$$

Then $\hat{\phi}_T$ concentrates near $\pm\gamma_0$ with width $\sim T^{-1/2}$; let $\phi_T = \mathcal{F}^{-1}\hat{\phi}_T$, with $\|\phi_T\|_2 = 1$.

2.1.2 Frame capture and discrete-to-continuous limit

By T2, $\sum_{\xi \in \Xi_A} w_{p,k} |\hat{\phi}_T(\xi)|^2 \geq C_{\text{fr}}^{(w)}(A)$. Moreover:

Lemma 2.1 (Frame-to-integral)**.**

$$\sum_{\xi \in \Xi_A} w_{p,k} |\hat{\phi}_T(\xi)|^2 \;\longrightarrow\; \int_{-A}^{A} |\hat{\phi}_T(\gamma)|^2 d\gamma \;=\; 1 \qquad (T \to \infty).$$

2.1.3 Archimedean term $\to 0$ (explicit DCT)

Write $A(\phi) = \int K(t)\hat{\phi}(t)dt$ with $K(t) = 2\log|t| + O(|t|^{-2})$. Since $\operatorname{supp}(\hat{\phi}_T) \subset \{|t - \gamma_0| \leq c/\sqrt{T}\}$ and $\|\hat{\phi}_T\|_1 = O(1)$, choose

$$h(t) = C\,(1+t^2)^{-1}(1 + \log^2(2+|t|))\,\mathbf{1}_{|t-\gamma_0|\leq 2}(t) \in L^1,$$

and note $|K(t)\hat{\phi}_T(t)| \leq h(t)$ for T large; as $\hat{\phi}_T(t) \to 0$ a.e., DCT gives $A(\phi_T) \to 0$.

2.1.4 Prime–power side $\to 0$ (no square)

The explicit formula involves $\sum_{p,k}(\log p)/p^{k/2}\ \phi_T(\xi_{p,k})$ (not squared). By Fourier inversion and repeated integration by parts, for any $M \in \mathbb{N}$ there exists C_M with

$$|\phi_T(x)| \le C_M \frac{T^{(M-1)/2}}{(1+|x|)^M}.$$

For fixed A there are finitely many $\xi_{p,k} \in [0, A]$, hence

$$\sum_{p,k} \frac{\log p}{p^{k/2}} \left|\phi_T(\xi_{p,k})\right| \ \le\ \max_{\xi\in\Xi_A} |\phi_T(\xi)| \sum_{p,k:\ \xi_{p,k}\le A} \frac{\log p}{p^{k/2}} \ \xrightarrow[T\to\infty]{}\ 0,$$

since $\max_{\xi\in\Xi_A} |\phi_T(\xi)| \to 0$ (width $T^{-1/2}$, amplitude $\to 0$ pointwise).

2.1.5 Contradiction

The explicit formula yields

$$\sum_{\rho} \hat{\phi}_T(\gamma_\rho) = A(\phi_T) + \sum_{p,k} \frac{\log p}{p^{k/2}} \phi_T(\xi_{p,k}).$$

Letting $T \to \infty$, LHS$\to 1$, RHS$\to 0$; contradiction.

A Guinand–Weil explicit formula

For $\phi \in \mathcal{PW}$ even, smooth, band-limited, contour-shift and residues for the completed ξ–function yield

$$\sum_{\rho} \hat{\phi}(\gamma_\rho) = \int K(t)\, \hat{\phi}(t)\, \frac{dt}{2\pi} + \sum_{p,k\ge 1} \frac{\log p}{p^{k/2}} \phi(k \log p),$$

with $K(t) = \psi(\frac{1}{4} + \frac{it}{2}) + \psi(\frac{1}{4} - \frac{it}{2}) - 2\psi(\frac{1}{4})$ and $\psi = \Gamma'/\Gamma$.

B Archimedean control (IBP)

Using $\psi(\frac{1}{4} \pm \frac{it}{2}) = \log|t| + O(|t|^{-2})$ uniformly and a $\cos^2$ taper with plateau+transition, two integrations by parts on the boundary layer give

$$|A(\phi)| \ \le\ \frac{C}{A^2}\, \|\hat{\phi}\|_1.$$

C Absorption lemma

If $a \le \eta a + b$ with $\eta \in (0, 1)$, then $a \le b/(1-\eta)$.

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